

How the enquiry engine works

Draft for discussion, 7 October 2026. RSTN runs the system: intake, case records, status tracking and the staff screen. Synvo provides the engine behind it as a hosted API, reached through a privacy connector inside RSTN's network: one call per email returns the issues found, a recommended action for each, a source-cited and independently checked reply, and a full decision trace. PaCE staff review and carry out every action.

Workflow and architecture

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Solved, shown in POC

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Solved, needs NTU data source

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Partly solved

●

New build

●

RSTN / NTU scope

AI

AI reasoning step

rules

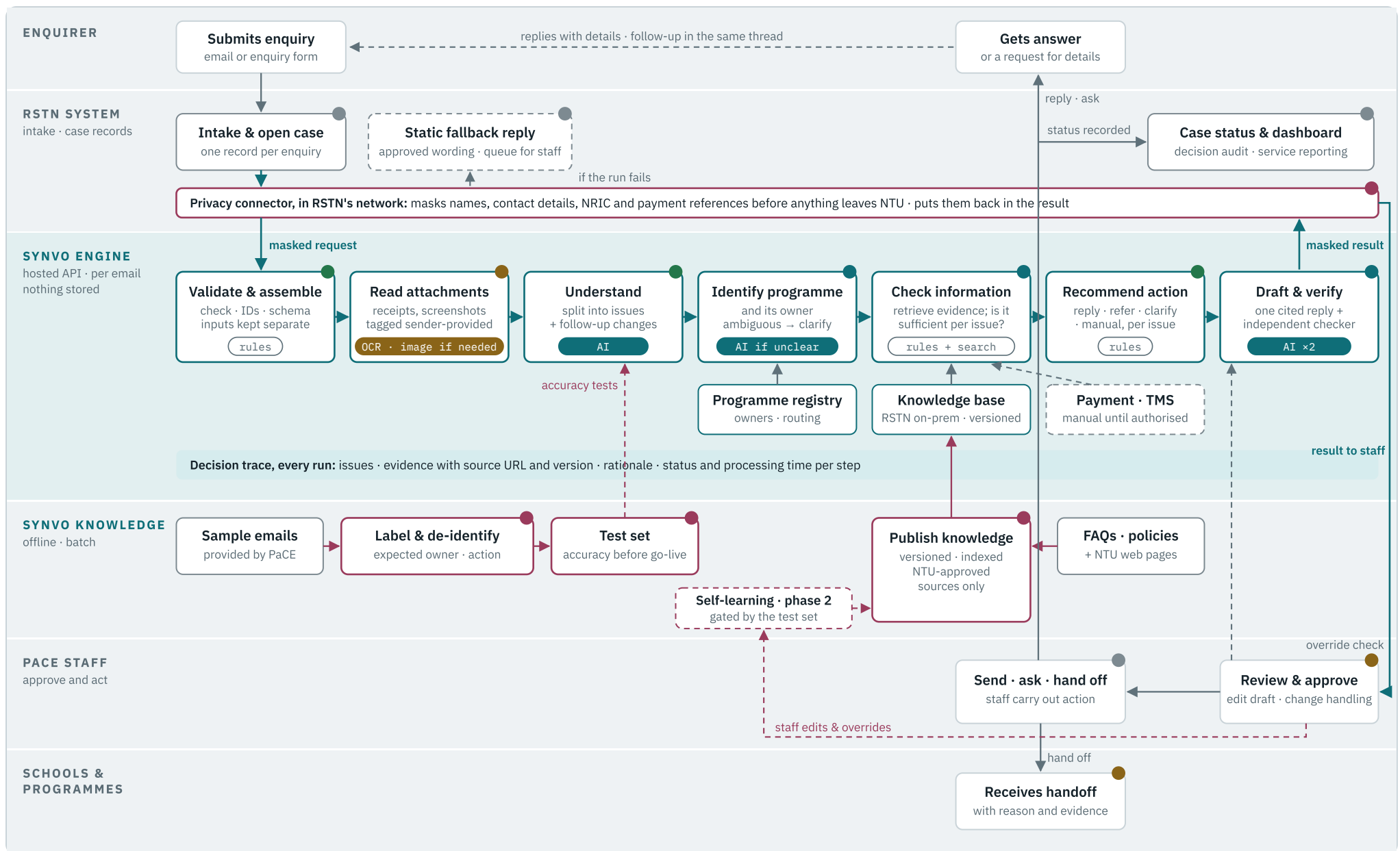
fixed, testable rules

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solid: happens for every enquiry

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dashed: only in some cases, or phase 2



- RSTN calls the engine once per email. When staff change how an issue is handled, RSTN sends the email and the previous result again with the change, and the engine re-checks only that issue. Nothing is held between calls. An invalid change is rejected with a reason and the plan stays as it was.

- Requests and results pass through the privacy connector inside RSTN's network, so the engine only ever sees placeholders such as [NAME_1].
- If the engine fails at any step, nothing it produced reaches the enquirer. RSTN sends its approved acknowledgement and queues the case for staff.
- Sample emails from PaCE are used as a test set and to find knowledge gaps. They are never used to train a model or cited as a source of facts.

The seven steps, on one example

The POC example: *"I'm looking at the data science SCTP. Is the full-time intake still open? Does the programme provide a training allowance? I have attached my receipt. Can you confirm that my payment has been received?"* with a bank-transfer screenshot attached.

STEP	WHAT IT DOES	ON THE EXAMPLE
Validate & assemble	Checks the request; keeps sender text, earlier thread messages and NTU data apart, so a sender's claim is never treated as an NTU fact.	One email, one attachment, no thread history.
Read attachments	Reads receipts and screenshots. The result is marked as sender-provided evidence.	Reads date, amount, payee and reference. Proves a receipt was sent, not that NTU received payment.
Understand	Splits the email into separate issues. On a follow-up, finds what is new, corrected or newly supplied.	Three issues: intake, training allowance, payment.
Identify programme	Matches each issue to a programme and its owning team in the registry. Similar names return a shortlist instead of a guess.	Advanced Professional Certificate in Data Science and AI, owned by PaCE.
Check information	Retrieves approved evidence and decides whether it answers the issue: sufficient, insufficient, conflicting or out of date.	Intake and allowance: sufficient (programme page). Payment: needs the payment system.
Recommend action	Picks one action per issue with fixed rules.	Reply directly, reply directly, manual handling.
Draft & verify	Writes one reply for the answerable issues, each claim linked to its source; rule checks and a separate checker block any unsupported statement.	One reply covering intake and allowance, with links to the NTU page.

What RSTN sends and what comes back

Field meanings below are proposed; the exact XML schema is to be agreed with RSTN during integration.

Request, one per email

Request

- └ requestId, correlationId, idempotencyKey
- └ Message senderRef, subject, body, channel,
| receivedAt, threadId, inReplyTo
- └ Thread[] permitted earlier messages
- └ Attachments[] type, reference or inline data
- └ NTUContext[] optional, authorised source only

Result

Result

- └ runHealth SUCCEEDED | DEGRADED | FAILED
- └ summary issue count, count per action
- └ issues[] programme, owner, routing basis,
| action, reason, evidence[] (URL, version)
- └ reply text, issues covered, claims → sources,
| SAFE_TO_REVIEW | BLOCKED | NOT_PRODUCED
- └ handoffNotes[] for the receiving team
- └ trace per step: status, time

ACTION	SHOWN TO STAFF AS	WHEN
ANSWER	Reply directly	Approved information answers the issue
HANDOFF	Refer to receiving team	Another school or team owns it, e.g. FlexiMasters in IC Design · ADMISSION → School of EEE
CLARIFY	Ask for clarification	Programme, intake or the actual question is missing or ambiguous
MANUAL_REVIEW	Manual handling	Needs NTU system data (payment, application, TMS) or a human judgement

The engine recommends; RSTN's workflow records what actually happened. These stay distinct in the data:

HANDOFF ≠ forwarded

SAFE_TO_REVIEW ≠ approved

MANUAL_REVIEW ≠ reviewed

reply draft ≠ sent email

How sample emails are used

No model is trained on NTU emails. The engine answers only from NTU-approved sources: programme pages, FAQs and policies. Sample emails from PaCE are used to test and tune it:

USE	HOW
Test set	Each sample is de-identified and labelled with PaCE: the right programme, owner and action, and what a good reply covers. Accuracy is measured on this set before go-live and after every change.
Gap list	Questions that come up in samples but no approved source answers are listed for NTU, so FAQs and pages can be added before go-live instead of those emails falling to manual handling.
Reply style	A few approved replies show PaCE's tone and structure. They guide wording only and are never cited as a source of facts.

What helps most is a few hundred de-identified emails covering the main programmes, multi-question emails, follow-ups and attachments, together with how PaCE actually handled each one (answered, forwarded and to whom, or asked for details).

How it is delivered

Synvo hosts the engine in Singapore. A small privacy connector, which Synvo supplies, runs inside RSTN's network. Every request passes through the connector, which replaces personal data with placeholders before the email leaves NTU and puts it back in the result before staff see it. The engine itself never sees who the enquirer is.

PROPOSED

Hosted engine + privacy connector

Names, contact details, NRIC and payment references stay inside NTU. RSTN runs one small container, with no GPU and no business logic. Engine and quality improvements arrive without a redeploy. Priced per email processed.

IF THE CONNECTOR CANNOT BE HOSTED

Hosted engine only

RSTN calls the engine directly; personal data is masked on arrival and nothing is kept after the response.

ONLY IF POLICY REQUIRES

On-premise licence

The whole engine runs inside NTU's network on local hardware. Higher fixed cost and slower updates; quoted separately.

Approved knowledge (programme pages, FAQs, policies) is synced to the engine as a versioned copy, with an immediate update when content is withdrawn. If a copy is not allowed, RSTN can expose a query interface instead.

Protecting enquirer data

Enquiry emails contain personal data. The engine is designed so that as little of it as possible leaves RSTN, is exposed during processing, or is kept anywhere.

CONTROL	WHAT IT MEANS
Masked inside NTU	The privacy connector in RSTN's network reads attachments with OCR and replaces names, email addresses, phone numbers, NRIC/FIN and payment references with placeholders before the request leaves NTU. It restores them in the result on RSTN's side. The mapping never leaves NTU.
No retention, no training	Enquiry content is never used for training and is not retained by Synvo or by any service involved in processing. Processing runs in Singapore.
Nothing kept after the response	The engine processes each email and returns the result. It keeps no copy of the email, attachments or reply. The decision trace holds IDs, labels, source references and timings, not email text.

CONTROL	WHAT IT MEANS
Encrypted and isolated	TLS with authenticated requests from RSTN; encryption at rest for anything stored, such as test data; a separate deployment and keys for this client.
Controlled access	Least-privilege staff access with multi-factor login; every access logged; no client data in development environments.
Only what is needed	The engine does not need the sender's email address, recipients or CC; RSTN passes an opaque sender reference instead.
Agreed in contract	A data processing agreement covering Synvo's role as data intermediary under the PDPA, purpose, retention and breach notification.

Masking does not reduce accuracy in principle: the engine decides from the programme and the question, not from who is asking. This will be confirmed on the sample-email test set before go-live.

What drives cost per email

- **Little AI work per email.** Only two of the seven steps always use AI reasoning; two more use it only when needed (attachments, unclear programme), and three are fixed rules. Most of the AI work is in drafting and checking the reply.
- **Delivery model.** The hosted API is priced per email processed; an on-premise licence is mostly fixed cost and quoted separately.
- **How much evidence is retrieved.** Drafting and checking grow with the size of the source excerpts, so evidence per issue is capped.
- **Attachments.** Text-heavy receipts can be read cheaply with OCR; app screenshots may need image understanding, which costs more. Sample screenshots will settle this.
- **Measured, not assumed.** Every result records processing cost and time per step. Before go-live, a shadow run on live traffic, with nothing sent, gives cost and latency per email by type (simple, multi-question, follow-up, with attachment).

What we need from RSTN and NTU

ITEM	WHY
Agreement to sync a versioned copy of approved knowledge to the hosted index, or a query interface; how versions and withdrawals are exposed	Every cited fact must point to a current, approved source
Data policy: acceptance of the hosted engine with the privacy connector (only masked text leaves NTU, processed in Singapore, nothing retained); residency or data-classification rules	Confirms the delivery model
A host for the privacy connector in RSTN's network (one small container, no GPU)	Masking before data leaves NTU
PDPA requirements for a data intermediary, DPA template, any security assessment needed	Data processing agreement
Programme registry and routing directory: programmes, aliases, owning teams, inboxes; who maintains them	Programme and owner identification, referrals
De-identified sample emails, with how PaCE handled each one	Accuracy test set and knowledge gap list
XML schema or sample payloads, thread identifiers, attachment handling	Freeze the API contract
10–30 redacted sample screenshots and attachments	Choose how attachments are read
Peak volume, acceptable time per email, synchronous or callback	Sizing and deployment
Handling of failed or blocked runs; fallback acknowledgement wording	Failure path
Any authorised interface for payment, application or TMS status	Until then those issues stay manual handling
Acceptance measures and targets: routing accuracy, first-contact resolution, handling time	Phase 1 sign-off